Akoizo Original Practice - Form 3 - Answer key

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A missed reading should be entered as…

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Missing with a reason

B. Zero automatically

C. The expected mean

D. A convenient nearby value

Correct: A

2. (2 points) The best final coherence check compares…

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only handwriting

B. Only the materials title

C. Question, variables, data, graph, and conclusion

D. Only page colors

Correct: C

3. (2 points) An inconvenient point should…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Always be deleted

B. Always be replaced by the mean

C. Be hidden in the graph

D. Be reviewed transparently using evidence

Correct: D

4. (2 points) Repeated readings on one plant always equal independent plants?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. No

B. Yes

C. Only at noon

D. Only with a calculator

Correct: A

5. (2 points) An abstract should include…

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

A. Every raw value

B. Only a title

C. The principal result and bounded conclusion

D. Only future promises

Correct: C

6. (4 points) Calculate mean and range for 8,10,12.

answer: Mean 10; range 4 response units. (4 points).

7. (4 points) Give three controlled conditions.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Same object, surface, release method, travel distance, and timing method are examples. (4 points).

8. (4 points) Explain late stop timing direction.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Stopping after the true endpoint increases recorded duration. (4 points).

9. (4 points) Why not invent qualitative notes after analysis?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: They would not be genuine observations and could falsely justify selective exclusion or an unsupported mechanism. (4 points).

10. (4 points) What should a changed procedure record?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Version, affected trials, reason, and whether combining data remains justified. (4 points).

11. (4 points) What limits extrapolation?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Only the tested range and conditions were observed; the relationship may change outside them. (4 points).

12. (6 points) Solve this case and explain your reasoning: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

reasoning: Identify time and level as systematically associated.
Upward drift can imitate or exaggerate a positive level effect.
Randomize order or use suitable blocking and calibration checks. (3 points).

conclusion: Replication alone does not remove systematic confounding. (3 points).

13. (6 points) Solve this case and explain your reasoning: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

reasoning: Area before rounding is 6.00 cm^2; the limiting measured input has two significant figures.
Report area as 6.0 cm^2 under the usual multiplication rule.
For addition, 2.0 + 3.00 = 5.0 cm, retaining the limiting tenths place. (3 points).

conclusion: Multiplication and addition use different classroom precision rules. (3 points).

14. (2 points) Define “Mean” in the context of Experimental Design.

answer: Arithmetic average. (2 points).

15. (2 points) Define “Dependent variable” in the context of Experimental Design.

answer: Measured response. (2 points).

16. (2 points) Define “Systematic bias” in the context of Experimental Design.

answer: Consistent or structured distortion. (2 points).

17. (2 points) Define “Protocol deviation” in the context of Experimental Design.

answer: Departure from the stated procedure. (2 points).

18. (2 points) Define “Guard digits” in the context of Experimental Design.

answer: Extra intermediate digits retained to avoid rounding error. (2 points).